
UNIT 11 TRADITIONAL MODELS OF BUSINESS CYCLES*

Structure

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11.0 OBJECTIVES

After going through this unit you would be able to

- explain the concept and features of business cycle;
- identify the various phases of business cycle;
- ascertain the theoretical framework which explains the occurrence of business cycle;
- appreciate the Keynesian view on business cycle;
- explain the multiplier-accelerator model of business cycle given by Samuelson;
- examine the political business cycle theory;
- appreciate the methodology used by NBER to date the business cycle; and
- distinguish between the leading, lagging and coincident indicators.

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11.1 INTRODUCTION

Economic growth witnessed by a country is smooth; it passes through many ups and downs. There have been periodical ups and downs in the GDP levels of these countries. It is interesting to note that such fluctuation is not limited to economic growth only, it permeates various economic aggregates such as income, employment and prices around their long term trends. Most developed countries have experienced phases of expansion and contraction in output and other economic aggregates. These alternating phases of upswings and downswings are known as business cycles.

You should note that formal analysis of business cycles started in the early 20th century. It was observed that periods of expansion and contraction in an economy occur in a regular manner. The characteristics of these periods are documented by economists such as Wesley Mitchell, Simon Kuznets, and Frederick Mills. Mitchell documented the co-movement of variables over the cycles; Mills documented the co-movement of prices and quantities over expansions and contractions; Kuznets studied the patterns of both growth and fluctuations. The 1930s was a very active period in business cycle research. However, interest in business cycles declined after the publication of Keynes' General Theory which turned attention away from business cycles to short run management of the economy. Interest in business cycles revived in the 1970s when the prevalent economic crisis in many countries could not be explained by Keynesian model.

11.2 FEATURES OF BUSINESS CYCLES

Business cycles consist of recurrent alternating phases of expansions and contractions in a number of economic variables including employment, production, real income, and real sales. Business cycles involve multidimensional processes, in which quantities and prices, stocks and flows, outputs and inputs, real, monetary and financial variables all tend to move together. These are asymmetric in the sense that expansion typically exceeds contractions in size and duration. Business cycles can be distinguished from other fluctuations as they are usually *larger, longer, and widely diffused*.

The major features of business cycles are as follows:

- 1) Though business cycles do not show the same regularity, they have certain distinct phases such as expansion, recession, depression, and recovery. The duration of a business cycle can vary from two years to twelve years.
- 2) Business cycles are *synchronic*. Depression or contraction occurs simultaneously in most industries or sectors of the economy. Recession passes from one industry to another and chain reaction continues till the whole economy is in the grip of recession. Similarly, prosperity spreads through various linkages of input-output relations or demand relations between industries or sectors.

- 3) Fluctuations occur simultaneously in the level of output as well as employment, investment, consumption, rate of interest, price level, etc.
- 4) Consumption of durable goods and investment are affected the most by cyclical fluctuations. As stressed by Keynes, investment is very unstable as it depends on profit expectations of private entrepreneurs. Any change in these expectations makes investment unstable. Thus the amplitude of fluctuation in the case of durable household effects is higher than that of GDP.
- 5) Consumption of non-durable goods and services do not vary much during the different phases of business cycles. Past data of business cycles reveal that households maintain a great stability in the consumption of non-durable goods. Thus the amplitude of fluctuation in the case of non-durable consumption goods is lower than that of GDP.
- 6) The immediate impact of recession or expansion is on the inventories of goods. When recession sets in, inventories start accumulating beyond the desired level. It leads to cut in production of goods. In contrast, when recovery starts, aggregate demand picks up and inventories go below the desired level. It encourages business houses to place more orders for goods which boosts production and stimulates investment.
- 7) Profits fluctuate more than other types of income as business cycles cause uncertainty for the businessmen and make it difficult to forecast economic conditions. During depression, profits turn negative and many businesses go bankrupt.
- 8) Business cycles are international in character. That is, once started in one country, it spreads to other countries through contagion effect. The downslide in financial markets, for example, in one country spreads rapidly to other countries as financial markets are linked globally through capital flows. Further, recession in one country, say the United States (US), can spread to other countries as the imports of the US will decline. Countries which are major exporters to the US will witness a decline in their exports and may witness recession.

11.3 PHASES OF BUSINESS CYCLES

Business cycles are characterized by expansion of economic variables in one period and contraction in the subsequent period. Generally, there are four phases of a business cycle, viz., expansion, recession, depression, and recovery. The four phases of a business cycle are depicted in Fig. 11.1. In the figure the upward sloping straight line indicates the steady state growth path or the long run growth path of GDP. The GDP fluctuates around the steady state growth path due to business cycles. In the expansion phase there is acceleration in growth rate while in the recession phase there is a deceleration in growth rate. In the depression phase economic growth is below its long run trend and the economy can witness negative growth rate also. In fact, negative growth for an economy is not rare. In

2019, several countries (for example, Venezuela, Argentina and Turkey) are having negative economic growth. The difference between recession and depression is one of degree and extent. Similarly, the difference between recovery and expansion is one of degree and extent. After negative growth, the economy passes through the recovery phase and then through the expansion phase. The point at which an expansion ends and a recession begins is called the 'peak' of the business cycle. The point at which a depression ends and recovery begins is called the 'trough'. Thus peak and trough are the turning points in a business cycle.

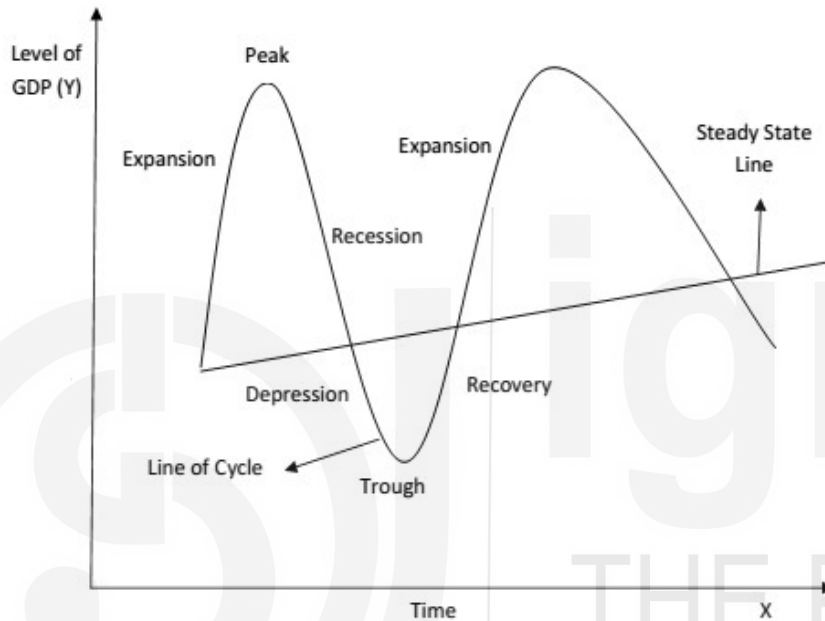


Fig. 11.1: Phases of a Business Cycle

11.3.1 Expansion Phase

In the expansion phase, there is an increase in various economic activities such as production, employment, output, wages, profits, demand and supply of products, and sales. An expansion phase can begin as a result of many forces, including willingness of financial institutions to lend more and willingness of business houses to borrow more. There is overall optimism in the economy. The expansion phase continues till the economic environment is favourable.

During the expansion phase, the economy often gets overheated in the sense that various constraints and frictions develop in the economy. Wage rate and prices increase much faster than output, leading to hike in production cost decline in profits. The central bank also pursues a restrictive monetary policy so that inflation is in control.

Economic growth in the expansion phase eventually slows down and reaches its peak. During the peak of a business cycle, economic variables such as production, profit, sales and employment are high; but do not accelerate further. There is a gradual decrease in the demand for various inputs due to the increase

in input prices. The increase in input prices leads to increase in product prices, while real income of people does not increase proportionately. It leads consumers to restructure their monthly budget and the demand for products, particularly luxuries and consumer durables, starts falling. The peak also occurs before various economic indicators, such as retail sales and the number of employed people, fall. When the decline in the demand for products becomes rapid and steady, 'recession phase' begins.

11.3.2 Recession Phase

In the recession phase, all economic variables such as production, prices, saving and investment start decreasing. Generally, in the beginning of the downturn, producers are not aware of the decrease in demand for their products and they continue to produce goods and services. In such a case, supply exceeds demand and there is accumulation of inventories.

Over time, producers realize that there is an unwanted accumulation of inventories, escalation in production cost, and decline in profits. Such a condition is first experienced by few industries and slowly spreads to the whole economy. During the recession phase, producers usually avoid new investments which lead to the reduction in the demand for factors of production, and consequent decline in input prices and unemployment. Firms reduce levels of production and the number of people on their payrolls. A chain reaction starts, lower income, lower demand, lower output, lower employment, lower income, and so on. The adverse effects of recession extend beyond the purely economic realm and influence the social fabric of society as well. Social unrest and crimes tend to rise during recessions.

During the recession phase, actual output may be below the potential output, thereby indicating a positive output gap. When recession continues further, economic growth rate may be negative also. In such cases, there is not just a decline the growth rate; absolute level of GDP declines. As sales decline, business houses find it difficult to repay their debts. As business sentiments are low enough to carry out new investment, demand for credit declines. Banks also become cautious in their lending, as the chances of default on repayment increases.

The economy however revives its growth rate over a period of time and optimism build up in certain sectors of the economy. This leads to reversal of the recession phase and the 'recovery phase' starts. Individuals and organizations start developing a positive attitude toward the various economic factors, such as investment, employment, and production. In the recovery phase there is an increase in consumer spending and demand for consumer goods. This provides incentives to firms to increase production, carry out new investments, etc. Further, there could be some investment during the recession phase due to replacement of obsolete machineries and maintenance of existing capital stock.

Price mechanism plays a very important role in the recovery phase of economy. As pointed out earlier, during the recession phase decline in input prices is greater than decline in product prices. This leads to a reduction in cost of production and increase in profits. Apart from this, in recovery phase, some of the depreciated capital goods are replaced by producers and some are maintained by them. As a result, investment and employment by organizations increases. As this process gains momentum an economy again enters into the phase of expansion. Thus, a business cycle gets completed.

Check Your Progress 1

1) What is meant by business cycle?

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2) List the important features of a business cycle.

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3) Describe the different phases of business cycle?

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11.4 THEORIES OF BUSINESS CYCLES

An important question is what causes business cycles. Several theories of business cycles have been propounded from time to time. Each of these theories spells out different factors which cause business cycles.

11.4.1 Keynes' Theory of Business Cycle

Keynes in his seminal work 'General Theory of Employment, Interest and Money' made an important contribution to the analysis of the causes of business cycles. According to Keynes, the changes in the level of aggregate effective demand will bring about fluctuations in the level of income. Aggregate demand comprises demand for consumption goods and demand for investment goods. While, propensity to consume being more or less stable in the short run, fluctuations in aggregate demand depend primarily upon fluctuations in investment demand. Multiplier plays a significant role in causing magnified

changes in income following a reduction or increase in investment. However, the theory fails to explain the cumulative character of business cycle. For example, suppose that investment rises by 100 rupees and that the magnitude of multiplier is 4. From the theory of multiplier, we know that national income will rise by 400 and if multiplier is the only force at work that will be the end of the matter, with the economy reaching a new stable equilibrium at a higher level of national income. But in real life, this is not likely to be so, for a rise in income produced by a given rise in investment will have further repercussions in the economy. This reaction is described in the principle of accelerator. Samuelson combined the accelerator principle with the multiplier and showed that the interaction between the two can bring about cyclical fluctuations in economic activity.

11.4.2 Samuelson's Model of Business Cycles: Interaction between Multiplier and Accelerator

Samuelson in his seminal paper convincingly showed that an autonomous increase in the level of investment raises income by a magnified amount depending upon the value of the multiplier. This increase in income further induces the increases in investment through acceleration effect. The increase in income brings about increase in aggregate demand for goods and services. To produce more goods we require more capital goods for which extra investment is undertaken. Thus the relationship between investment and income is one of mutual interaction; investment affects income which in turn affects investment demand and in this process income and employment fluctuate in a cyclical manner.

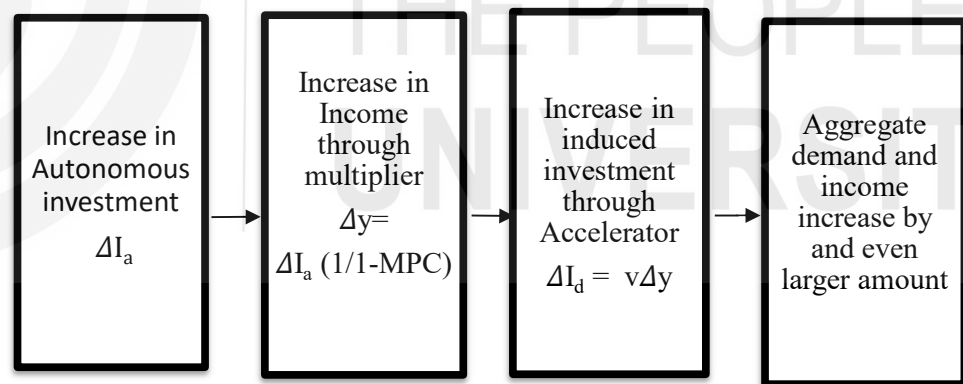


Fig. 10.2 shows how income and output will increase by even larger amount when accelerator is combined with the Keynesian multiplier.

ΔI_a = Increase in Autonomous investment

Δy = Increase in Income

$\frac{1}{1-MPC}$ = Size of Multiplier when MPC = Marginal Propensity to Consume

ΔI_d = Increase in Induced Investment

v = Size of accelerator

The changes in any component of aggregate demand produce a multiplier effect whose magnitude depends upon the marginal propensity to consume. When consumption, income and output increase under the influence of multiplier effect, they induce further changes in investment and the extent to this induced investment in capital goods industries depends on the capital output ratio, that is, the interaction between the multiplier and accelerator without any external shocks can give rise to the business cycles whose patterns differs depending upon the magnitudes of the marginal propensity to consume and capital-output ratio. The model of interaction between multiplier and accelerator can be mathematically represented as under:

$$Y_t = C_t + I_t \quad \dots (11.1)$$

$$C_t = C_a + c (Y_{t-1}) \quad \dots (11.2)$$

$$I_t = I_a + v (Y_{t-1} - Y_{t-2}) \quad \dots(11.3)$$

Where Y_t , C_t , I_t stand for income, consumption and investment respectively for a period t , C_a stands for autonomous consumption, I_a for autonomous investment, c for marginal propensity to consume and v for capital-output ratio or accelerator.

From equation 11.1, we notice that that consumption in a period t is a function of income of the previous period, Y_{t-1} . That is, one period lag has been assumed for income to determine the consumption of a period. Whereas, the induced investment in period t is a function of the change in income in the previous period in equation 11.2. This means that there are two period gaps for changes in income to determine induced investment. In the equation 11.3, induced investment equals $v (Y_{t-1} - Y_{t-2})$. Now, substitute equations 2 and 3 in equation 1 which gives us the equation 11.4.

$$Y_t = C_a + c (Y_{t-1}) + I_a + v (Y_{t-1} - Y_{t-2}) \quad \dots(11.4)$$

This equation reflects how changes in income are dependent on the values of marginal propensity to consume (c) and capital-output ration (v) or accelerator.

In static equilibrium, the level of income can be written as:

$$Y = C_a + cY + I \quad \dots(11.5)$$

This is due to the fact that in static equilibrium, given the data of the determining factors, the equilibrium level of income remains unchanged, that is, in this case, $Y_t = Y_{t-1} = Y_{t-2} = Y_{t-n}$ so that the periods lags have no influence at all and accelerator is reduced to zero as $Y_{t-1} - Y_{t-2} = 0$. Thus, in a dynamic state when autonomous investment changes, the equation 11.4 describes the path which a disequilibrium system follows to reach either a final equilibrium state or moves away from it. But whether the economy moves towards a new equilibrium or

deviates away from it depends on the values of marginal propensity to consume and capital-output ratio.

By taking different combinations of the values of marginal propensity to consume and capital-output ratio, Samuelson has described different paths which the economy will follow. The various combinations of the values of marginal propensity to consume and capital-output ratio are shown in Fig. 11.3.

Fig. 11.3: Combinations of c and v

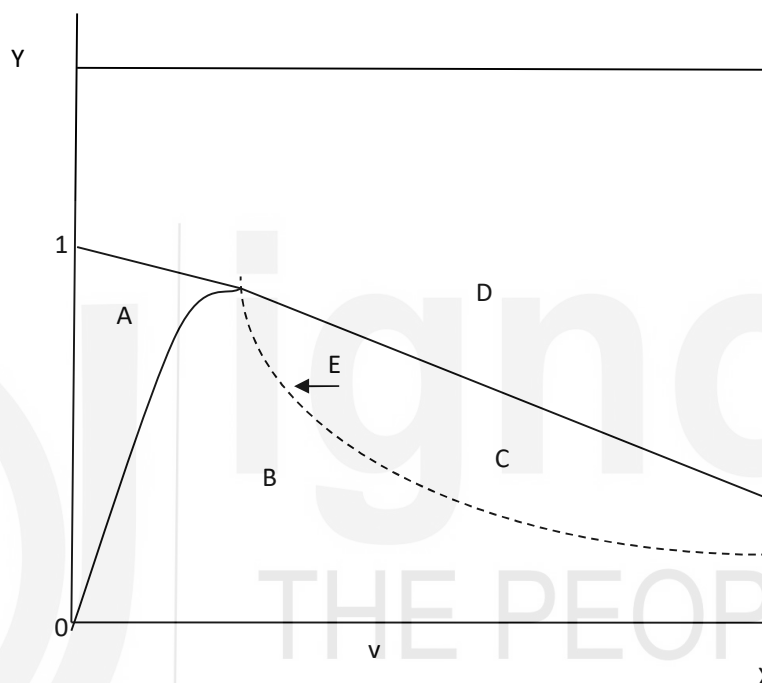
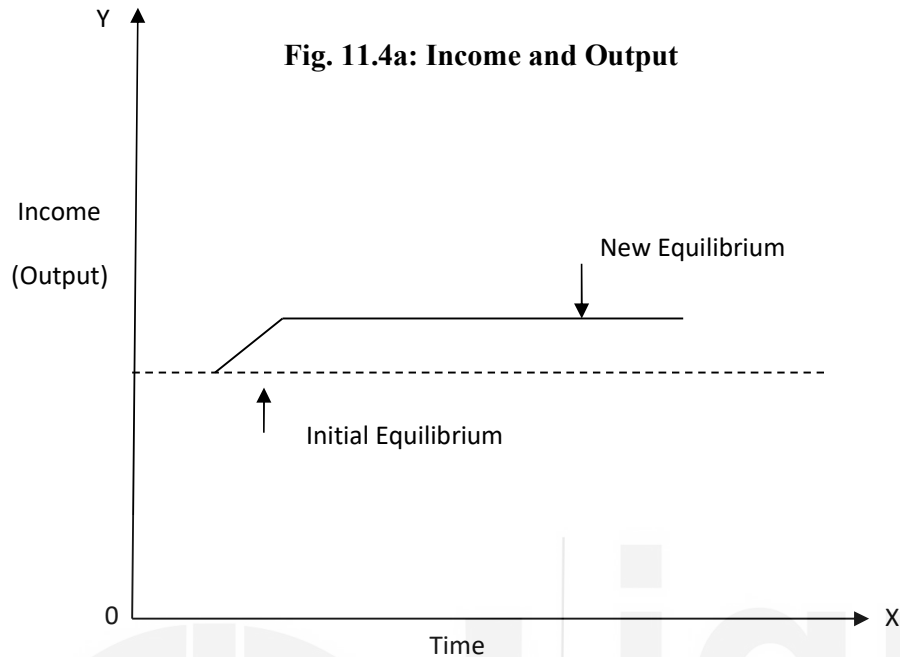


Fig. 10.3 shows the four paths which the economic activity can have depending upon combinations of the values of marginal propensity to consume (c) and capital-output ratio (v).

The four paths or patterns of movements which the economic activity (as measured by gross national product or income) can have depending upon various combinations of the values of marginal propensity to consume and capital-output ratio are depicted in Fig. 11.4 a to 11.4e. When the combinations of the value of marginal propensity to consume and capital-output ratio lie within the region marked as A, with a change in autonomous investment, the gross national product or income moves upward or downward at a decreasing rate and finally reaches a new equilibrium as is shown Fig. 11.4a.



If the values of c and v are such that they lie within the region B, the change in autonomous investment or autonomous consumption will generate fluctuations in income which follow the pattern of a series of damped cycles whose amplitudes go on declining until the cycles disappear as is shown in Fig. 11.4a.

The region C in Fig. 11.3 represents the combinations of c and v which are relatively high as compared to the region B and determine such values of multiplier and accelerator that bring about explosive cycles, that is, the fluctuations of income with successively greater and greater amplitude. The situation is depicted in Fig. 11.4c.

